

Gabriel Luz Almeida

Postdoctoral researcher in theoretical physics

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Research interests

I study the relativistic two-body problem and gravitational waves using effective field theory, post-Newtonian and post-Minkowskian methods, and multiloop techniques.

Appointments & education

2025- · Southeast University · Shing-Tung Yau Center
Postdoctoral researcher in theoretical physics · Nanjing, China
Advisor / supervisor: Zhengwen Liu

2023-2025 · USTC · PCFT / ICTS
Postdoctoral researcher · Hefei, China
Advisor / supervisor: Shuang-Yong Zhou

2019-2023 · UFRN · International Institute of Physics
PhD in Physics · Natal, Brazil
Advisor / supervisor: Riccardo Sturani

2017-2019 · Universidade Federal de Pernambuco
MSc in Physics · Recife, Brazil
Advisor / supervisor: Carlos Batista

2012-2017 · Universidade Federal de Pernambuco
BSc in Physics · Recife, Brazil

2015-2016 · Montana State University
Undergraduate exchange · Bozeman, USA

Awards

2024 · ICTP-SAIFR PhD Prize in Classical Gravity and Applications

2017 · Academic Laurel · Universidade Federal de Pernambuco

Journal articles & preprints

1. Angular momentum tail contributions to compact binary dynamics

Gabriel Luz Almeida, Alan Müller, Stefano Foffa, Riccardo Sturani
Physical Review D 114, 064057 (2026). [arXiv:2603.20096](#) · [DOI](#)

2. Conservative binary dynamics to third post-Minkowskian order beyond General Relativity

Gabriel Luz Almeida, Yuchen Du, Zhengwen Liu, Hongbin Wang
Preprint (2026). [arXiv:2602.08876](#)

3. Gravitational memory contributions to waveform and effective action

Gabriel Luz Almeida, Alan Müller, Stefano Foffa, Riccardo Sturani
Physical Review D 112, 084077 (2025). [arXiv:2410.10565](#) · [DOI](#)

4. Post-Newtonian dynamics of spinning black hole binaries in Einstein-scalar-Gauss-Bonnet gravity

Gabriel Luz Almeida, Shuang-Yong Zhou
Physical Review D 110, 124016 (2024). [arXiv:2408.14196](#) · [DOI](#)

5. Binary dynamics to second post-Newtonian order in scalar-tensor and Einstein-scalar-Gauss-Bonnet gravity from effective field theory

Gabriel Luz Almeida
Physical Review D 109, 084060 (2024). [arXiv:2402.13996](#) · [DOI](#)

6. Conservative binary dynamics from gravitational tail emission processes

Gabriel Luz Almeida, Alan Müller, Stefano Foffa, Riccardo Sturani
Physical Review D 108, 124010 (2023). [arXiv:2307.05327](#) · [DOI](#)
Erratum: [Physical Review D 113, 089903 \(2026\)](#)

7. Gravitational radiation contributions to the two-body scattering angle

Gabriel Luz Almeida, Stefano Foffa, Riccardo Sturani
Physical Review D 107, 024020 (2023). [arXiv:2209.11594](#) · [DOI](#)

8. Tail contributions to gravitational conservative dynamics

Gabriel Luz Almeida, Stefano Foffa, Riccardo Sturani
Physical Review D 104, 124075 (2021). [arXiv:2110.14146](#) · [DOI](#)

9. Gravitational multipole renormalization

Gabriel Luz Almeida, Stefano Foffa, Riccardo Sturani
Physical Review D 104, 084095 (2021). [arXiv:2107.02634](#) · [DOI](#)
Erratum: [Physical Review D 111, 129901 \(2025\)](#)

10. Classical gravitational self-energy from double copy

Gabriel Luz Almeida, Stefano Foffa, Riccardo Sturani
JHEP 11 (2020), 165 (2020). [arXiv:2008.06195](#) · [DOI](#)

11. A class of higher-dimensional solutions of Einstein's vacuum equation

Gabriel Luz Almeida, Carlos Batista
European Physical Journal C 79, 11 (2019). [arXiv:1808.09998](#) · [DOI](#)

12. A class of integrable metrics II

Carlos Batista, Gabriel Luz Almeida
Physical Review D 98, 044030 (2018). [arXiv:1805.09206](#) · [DOI](#)

13. Class of integrable metrics and gauge fields

Gabriel Luz Almeida, Carlos Batista
Physical Review D 96, 084003 (2017). [arXiv:1707.04630](#) · [DOI](#)

Theses & book

Classical amplitudes in gravitational-wave physics. PhD thesis · UFRN, 2023. [arXiv:2309.00772](https://arxiv.org/abs/2309.00772)

On symmetries of exact solutions of Einstein field equations. MSc thesis · UFPE, 2019.

Symmetries and exact solutions in general relativity. LAMBERT Academic Publishing, 2019. ISBN 978-6-200-48020-0.

Teaching

Introduction to Gravitational Wave Science · 2025–2026

Southeast University · Shing-Tung Yau Center

Co-initiator and instructor of a voluntary, non-credit course. I taught the first phase: twelve 90-minute lectures in English (18 hours), covering linearized gravity, gravitational-wave generation, compact-binary sources and an introduction to detection. The course uses Michele Maggiore's *Gravitational Waves*, Vol. 1.

NRGR — The PN-EFT for Compact Binaries · 27 – 28 Jan 2026

Advanced Lectures on Quantum Field Theory · Southeast University

Four 50-minute lectures on gravitational waves and Wilsonian EFT, scale separation and the method of regions, the radiation zone and matching, and nonlinear tails and multipole renormalization. The course connects self-energy diagrams to radiated flux through the optical theorem.

An EFT approach to GW physics · 19, 21 and 22 Sept 2023

PCFT / ICTS · University of Science and Technology of China

Three 90-minute lectures introducing NRGR and effective-field-theory methods for gravitational radiation and compact-binary observables.

Introduction to General Relativity · UFRN, 2022. Graduate teaching assistant; problem sessions and occasional regular lectures.

General Physics I · UFRN, 2021. Graduate teaching assistant; problem-solving classes.

Quantum Mechanics I · UFPE, 2017. Undergraduate teaching assistant; tutorials and problem sets.

Classical Mechanics I · UFPE, 2016. Undergraduate teaching assistant; tutorials.

General Physics IV · UFPE, 2014. Undergraduate teaching assistant; optics and introductory modern physics.

Selected talks

Pushing the Precision Frontier: Recent Developments in the EFT of Compact Binaries

Chengdu University of Technology, China · 22 May 2026

Tails of tails and the UV/IR pole splitting

QCD meets Gravity XI · ICTP-SAIFR, Brazil · 18 Dec 2025

EFT Techniques for Precision GW Predictions from Compact Binaries

Frontier Forum on Gravitational Waves · ICTP-AP / SEU, China · 27 May 2025

The Two-Body Problem in Beyond-General Relativity: Precision Modeling in Scalar-Tensor and ESGB Gravity

Effective Field Theories, Gravity and Cosmology · HIAS-UCAS, China · 28 Apr 2025

Conservative Binary Dynamics from Gravitational Tail Emission Processes

Gravitational Waves meet Amplitudes in the Southern Hemisphere · ICTP-SAIFR, Brazil · 21 Aug 2023

Gravitational Waves and the Relativistic Two-Body Problem from an Effective Field Theory Perspective

SISSA · Trieste, Italy · 7 Sept 2022

Research visits

Shing-Tung Yau Center · Southeast University · Dec 2024

Nanjing, China · research visit

ICTP-SAIFR / IFT-UNESP · Oct 2022 – Jun 2023

São Paulo, Brazil · long-term research visit

ICTP · HECAP · Sept 2022

Trieste, Italy · one-month visit

Université de Genève · Aug 2022

Geneva, Switzerland · ten-day visit

INFN · Padova · Jul 2022

Padua, Italy · one-week visit

Academic service

Referee for Physical Review D and Physical Review Letters.